

Program: Diploma in Renewable Energy	
Course Code: 6292C	Course Title: Sustainable Engineering
Semester: 6	Credits: 4
Course Category: Open Elective Course	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To have an increased awareness among students on the issues in areas of sustainability
- To understand the role of engineering and technology with sustainable development
- To know the methods, tools and incentives for sustainable product-service system development
- To establish a clear understanding of the role and impact of various aspects of engineering on environmental, societal and economic problems

Course Prerequisites:

Topic	Course Code	Course name	Semester
Work,Power and Energy		Applied Physics	1

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Outline the relevance and the concept of sustainability and the global initiatives in this direction	15	Understanding
CO2	Explain the different types of environmental pollution problems and their sustainable solutions	14	Understanding

CO3	Summarize the environmental regulations and standards	15	Understanding
CO4	Explain the broad perspective of sustainable practices	14	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2				3		
CO2	2				3		
CO3	2				3		
CO4	2				3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Outline the relevance and the concept of sustainability and the global initiatives in this direction		
M1.01	Outline the evolution of the concept sustainability	4	Understanding
M1.02	Summarize sustainable development, nexus between technology and sustainable development	2	Understanding
M1.03	Illustrate Millennium Development Goals (MDGs) and Sustainable Development Goals (SDGs)	4	Understanding
M1.04	Explain Clean Development Mechanism (CDM)	5	Understanding

Contents: Sustainability - Introduction, Need and concept of sustainability, Social, environmental and economic sustainability concepts.

Sustainable development, Nexus between Technology and Sustainable development.

Millennium Development Goals (MDGs) and Sustainable Development Goals (SDGs), Challenges for Sustainable Development. Multilateral environmental agreements and Protocols

Clean Development Mechanism (CDM), Environmental legislations in India - Water Act, Air Act.

CO2	Explain the different types of environmental pollution problems and their sustainable solutions		
M2.01	Explain air pollution and water pollution – sources and effects	3	Understanding
M2.02	Illustrate zero waste concept and 3 R concepts in solid waste management	3	Understanding
M2.03	Outline greenhouse effect, global warming, climate change, ozone layer depletion	4	Understanding
M2.04	Explain legal provisions for environmental protection	4	Understanding
	Series Test – I	1	

Contents:

Air Pollution, Effects of Air Pollution; Water pollution- sources

Sustainable wastewater treatment, Solid waste - sources, impacts of solid waste, Zero waste concept, 3 R concept.

Global environmental issues- resource degradation, climate change, global warming, ozone layer depletion.

Regional and local environmental issues, carbon credits and carbon trading, carbon foot print ,legal provisions for environmental protection

CO3	Summarize the environmental regulations and standards		
M3.01	Explain environmental management standards	4	Understanding
M3.02	Summarize ISO 14001:2015 frame work and benefits	5	Understanding
M3.03	Outline scope and goal of Life Cycle Analysis (LCA)	3	Understanding

M3.04	Explain Environment Impact Assessment (EIA)	3	Understanding
Contents: Environmental management standards ,ISO 14000 series, Benefits of ISO 14000 series, ISO 14000 Frame work Life Cycle Analysis (LCA) - Components, LCA Process, Product life cycle, Procedure for LCA, Limitations Life Cycle Impact Assessment; Environmental Impact Assessment - Objectives, EIA Process & Procedures, Benefits of EIA, Procedures of EIA in India			
CO4	Explain the broad perspective of sustainable practices		
M4.01	Outline basic concept of sustainable habitat	2	Understanding
M4.02	Summarize methods for increasing energy efficiency of buildings	4	Understanding
M4.03	Explain Green Engineering	4	Understanding
M4.04	Summarize Sustainable Urbanisation, Sustainable cities, Sustainable transport	4	Understanding
	Series Test – II	1	
Contents: Sustainable habitat - Basic concept Energy efficient buildings- Methods to increase efficiency Green Engineering- Basic concept - Benefits of Green Building , Green building certification, Different green building programmes. Sustainable Urbanisation, industrialisation and poverty reduction; Social and technological change, Industrial Processes: Material selection, Pollution Prevention, Industrial Ecology, Industrial symbiosis.			

Text / Reference

T/R	Book Title/Author
T1	Suresh K, A Text Book of Sustainable Engineering, Cheff Educational Publishers, Trivandrum, ISBN - 9788193167007
T2	Bradley. A.S; Adebayo,A.O., Maria, P. Engineering applications in sustainable design and development, Cengage learning
T3	Environment Impact Assessment Guidelines, Notification of Government of India, 2006.
T4	Mackenthun, K.M., Basic Concepts in Environmental Management, Lewis Publication, London, 1998
T5	ECBC Code 2007, Bureau of Energy Efficiency, New Delhi Bureau of Energy Efficiency Publications-Rating System, TERI Publications - GRIHA Rating System
R1	Ni bin Chang, Systems Analysis for Sustainable Engineering: Theory and Applications, McGraw-Hill Professional.
R2	Twidell, J. W. and Weir, A. D., Renewable Energy Resources, English Language Book Society (ELBS)

Online Resources

Sl.No	Website Link
1	https://onlinecourses.nptel.ac.in/noc21_ce47/preview
2	http://www.nitttrc.edu.in/nptel/courses/video/107103081/L01.html
4	https://www.coursera.org/learn/sustainability
5	https://www.keralanotes.com/2021/12/KTU-S3-Sustainable-Engineering-Notes.html?m=1#
6	https://www.ktunotes.in/introduction-to-sustainable-engineering-notes/